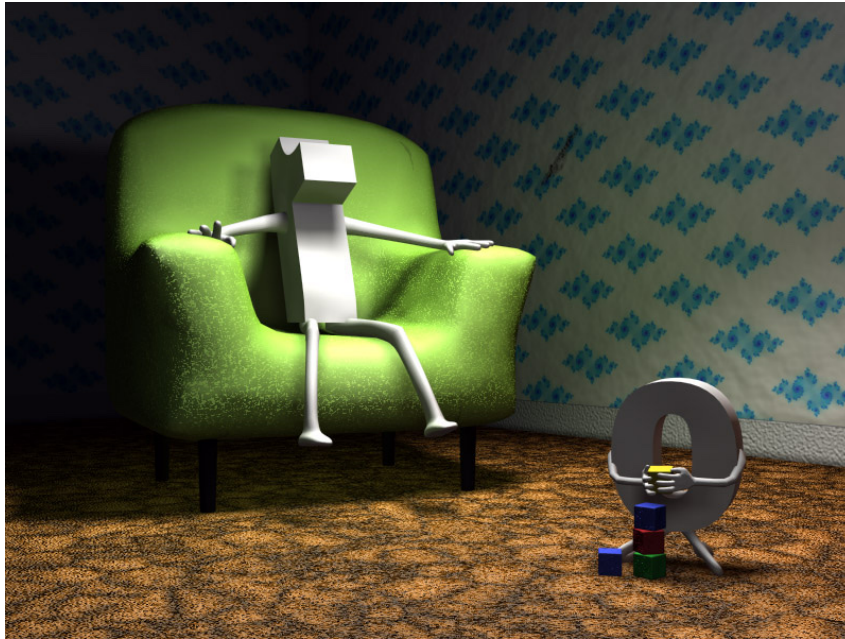


Chapter 1. INTEGERS

1. 2 Divisibility and Prime Factorisation

1. 2. 3 Euclidean Algorithm



Today we will learn a method to compute $\gcd(m, n)$ developed by Euclid. It is known as the

EUCLIDEAN ALGORITHM.

key result!

PROPOSITION 1.18. If $\underline{m} = q\underline{n} + r$ then $\gcd(m, n) = \gcd(n, r)$.

Proof. Let $d = \gcd(m, n)$ and $k = \gcd(n, r)$. By definition k is a common n and r , so k is a divisor of any linear combination of n and r (Theorem 1.10 (4)). In particular,

$$k \mid (qn + r), \text{ that is, } k \mid m.$$

We have that $k \mid m$ and also $k \mid n$. But then $k \mid d$ (because $d = \gcd(m, n)$).

Using now the linear combination $r = (-q)n + m$ and repeating a similar argument we have that $\underline{d} \mid \underline{k}$.

$$k \mid d \text{ and } d \mid k \xRightarrow[\text{Thm 1.10 (3)}]{d, k \geq 1} \underline{d = k} \quad \blacksquare$$

EUCLIDEAN ALGORITHM. Let m and n be integers, not both zero. We apply the *Division Algorithm* repeatedly:

$$\begin{array}{lll} m = q_1n + r_1 & \text{divisor: } n, & \text{remainder: } r_1 \\ n = q_2r_1 + r_2 & \text{divisor: } r_1, & \text{remainder: } r_2 \\ r_1 = q_3r_2 + r_3 & & \\ \vdots & & \end{array}$$

At each stage we divide the divisor at the previous stage by the remainder, so the **remainders form a decreasing sequence of nonnegative integers**:

$$n > r_1 > r_2 > r_3 > \dots \geq 0.$$

It is clear that (in at most n steps) we will reach a remainder of 0. Assume that r_t is the last nonzero remainder, then the last two equations are:

$$r_{t-2} = q_tr_{t-1} + r_t, \quad \text{and} \quad r_{t-1} = q_{t+1}r_t + 0.$$

Now, Proposition 1.18 gives

$$\gcd(m, n) = \gcd(n, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_{t-1}, r_t) = r_t$$

Thus $\gcd(m, n)$ is the **LAST NONZERO REMAINDER**.

Moreover, we can **express** $\gcd(m, n) = r_t$ as a **linear combination of m and n** (finding the coefficients in *Bézout's Lemma*) by eliminating the remainders $r_{t-1}, r_{t-2}, \dots$, successively from these equations.

Example 1.20. Find $\gcd(268, 112)$ and express it as a linear combination of 268 and 112.

$$268 = 2 \cdot \underline{112} + \underline{44} \quad (1)$$

$$112 = 2 \cdot \underline{44} + \underline{24} \quad (2)$$

$$44 = 1 \cdot \underline{24} + \underline{20} \quad (3)$$

$$24 = 1 \cdot \underline{20} + \underline{4} \quad (4)$$

$$20 = 5 \cdot 4 + \underline{0}$$

$$\gcd(268, 112) = 4 \quad (\text{the last nonzero remainder})$$

$$\begin{aligned} \underbrace{4}_d &= 24 + (-1) \cdot 20 \stackrel{(3)}{=} 24 + (-1)(44 + (-1)24) \\ &= \underline{2 \cdot 24} + (-1)44 \stackrel{(2)}{=} 2 \cdot (112 + (-2) \cdot 44) + \\ &\quad + (-1) \cdot 44 \end{aligned}$$

$$= 2 \cdot 112 + (-5) \cdot 44 \stackrel{(1)}{=}$$

$$= 2 \cdot 112 + (-5) (268 + (-2) \cdot 112) =$$

$$= \underbrace{(-5)}_x \cdot \underbrace{268}_m + \underbrace{12}_y \cdot \underbrace{112}_n$$

